

GridMind

Capacity planning for high-density data centers

Every new workload has to clear power, cooling, floor space and budget before it can be placed. Today four teams check those four things separately, over several days. GridMind checks them together and returns a plan the site can actually build.

Track

**The Fortified
Enterprise Fleet**

Live

[gridmind-wuvfpvopoq-uk.a.run.app](#)

Built on

Gemini 3.5 on Vertex AI
Cloud Run, Firestore

The problem

Four teams, four inboxes, and nobody who checks the four answers together



Power engineering

breaker capacity, spare circuits, grid events



Cooling and thermal ops

heat removal, efficiency, water permits



Facilities and DCOps

rack space, floor loading, install crews



Finance and procurement

budget, cost per GPU-hour, capex

**Each one is right.
Nobody owns the joint check.**

Requests move from team to team by email and spreadsheet. Every approval is correct on its own axis, and no one asks whether the four approvals describe the same plan.

What that leaves behind has a name in the industry: stranded capacity. Megawatts a site has paid for and cannot use.

days

for four sequential approvals

0

teams who check them together

~90 sec

for the same four checks, run together

What GridMind is worth

For the operations team that has to place a workload, GridMind returns one plan that clears power, cooling, space and budget at the same time, in about ninety seconds instead of several days, with a report showing what every assessor was shown beside what it concluded.

Today

- Four approvals, one team at a time
- Each team sees only its own data
- Nobody computes the joint check
- Conflicts surface after the racks arrive
- The reasoning lives in someone's inbox

With GridMind

- Four checks run together, in parallel
- The orchestrator sees all four verdicts
- It computes what all four constraints allow
- Conflicts surface before anything is ordered
- Every decision ships with its evidence

What GridMind does

A capacity request goes in. A plan the site can build comes out.

1

Submit a request

Rack count, power draw per rack, cooling type, weight, budget ceiling. A form, not free text.

2

Four assessors run in parallel

Power, cooling, facilities and cost. Each one reads only its own data and answers one question.

3

The orchestrator reconciles

It checks whether the four answers describe one buildable plan. When they do not, it opens a negotiation round.

4

You get a decision and a report

Target zone, the plan to get there, and what every assessor said beside the numbers it was shown.

About ninety seconds end to end. The same four checks take a site several days today.

A real run, round one

Six racks of GB200 NVL72. 792 kW, liquid cooled, 1,360 kg per rack.

Power

zone-a

most electrical headroom

Cooling

zone-c

only zone that can remove the heat

Facilities

zone-b

only zone with 6 free liquid racks

Cost

zone-b

avoids a \$48k per rack retrofit

Four correct answers. Three different rooms. No plan.

A system that counted votes would see four approvals and place the order. Then the racks arrive and the room cannot cool them.

This is the failure we built for. Not assessors contradicting each other, but assessors all being right and still adding up to nothing.

How the disagreement gets resolved

Bounded negotiation, and an intersection no single assessor could compute

Round 1

Four independent verdicts

conflict: zone_mismatch

Each assessor reads only its own data and cannot see what the others said.

Round 2

Positions attached

consistent, ends here

Each assessor is asked again with the others' positions attached. Facilities volunteers a retrofit it had not mentioned. Cost re-prices one rack, not six.

Round 3

The hard limit

3 rounds, then a human

If the verdicts still do not describe one plan, it stops and escalates with the whole disagreement attached. It never picks a winner.

Conflict detection is ordinary code, not a model and not a vote. Whether two zone ids differ is not a judgement call.

Each assessor also reports what it rules out

Cooling	zone-a	zone-b	zone-d
Power	zone-b	zone-d	
Facilities	zone-a	zone-d	
Cost	zone-a	zone-d	

survives

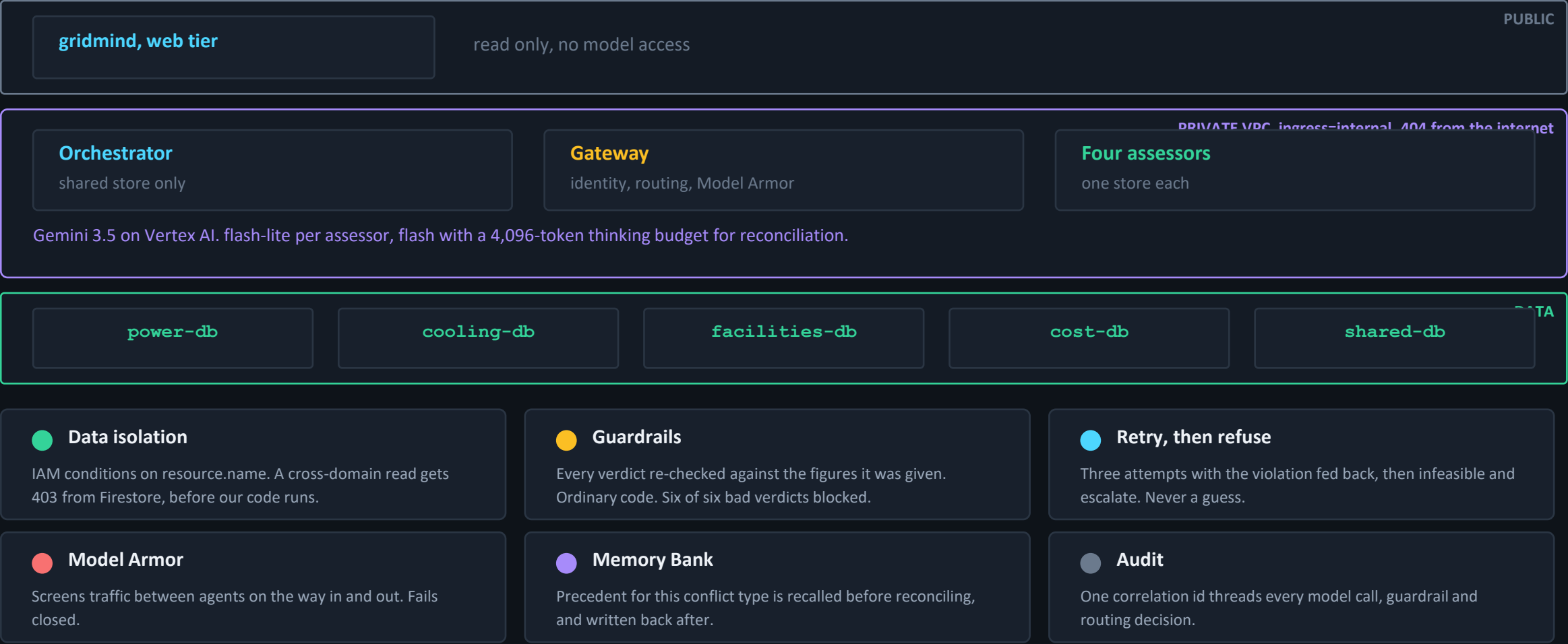
zone-c

In round one, not one assessor had picked zone-c. It only appears when you intersect four independent exclusion lists, which is exactly what no team can do from inside its own silo.

Final plan: zone-c, five liquid-ready racks plus one retrofit, fourteen hours of delay, \$48,000 rather than \$288,000.

Under the hood

Seven Cloud Run services, seven identities, five databases



It is deployed, and you can check it yourself

Every denial below is produced by Google Cloud, not by our application code

Run these against the live system

```
bash infra/iam/03_verify_isolation.sh      14 isolation checks
bash scripts/demo_denial.sh                7 live denials, every layer
python -m scripts.demo_guardrails          6 guardrail assertions
```

The orchestrator is incapable of reading facility data. That is a platform property.

Cloud Run services **7**

Security checks passing **21**

GEAP capabilities built **6 of 7**

Stated plainly

The facility is simulated. The agents, the infrastructure, the isolation and every denial shown are real. The data center is generated from published engineering figures, physically consistent, reproducible from a fixed seed.

How these decisions actually get made inside an operations team is something we would need to learn by shadowing one, not by writing a seed file. That is the honest boundary of what we built.

Four correct answers still add up to nothing unless something checks them together.

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Live

gridmind-wuvfpvopoq-uk.a.run.app

Code

github.com/PRACHIT27/GridMindAI

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